

A Deterministic Bandwidth-Mapper Scheme in Upstream TDM-PON for Time-Sensitive Industrial AI Applications

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Abstract *This paper proposes a real-time, scalable deterministic bandwidth-mapper (BWmapper) for upstream TDM-PON, capable of supporting a large number of Time-Sensitive Industrial-AI flows with diverse latency and jitter requirements alongside best-effort traffic. Validated through simulation and experiments, the scheme demonstrates low computational complexity and high efficiency. ©2026 The Author(s)*

Introduction

Industrial networks are increasingly required to support time-sensitive industrial Artificial Intelligence (AI) applications such as closed-loop control with learning-based controllers, vision-guided robotics, digital twins, and predictive maintenance. Unlike conventional IT or cloud AI workloads, industrial AI is tightly coupled to real-time physical processes. Its correct operation depend not only on throughput but, more critically, on deterministic packet delivery with bounded latency and jitter [1], since timing violations can destabilize control loops or invalidate inference results.

Time Division Multiplexing Passive Optical Network (TDM-PON) is a widely deployed fiber access technology that offers high capacity, scalability, and a cost-efficient point-to-multipoint architecture, making it attractive for industrial deployments. In TDM-PON, upstream (US) transmissions are controlled by a Dynamic Bandwidth Assignment (DBA) mechanism at the Optical Line Terminal (OLT), which allocates transmission opportunities to Optical Network Units (ONUs) in a time-division multiple-access (TDMA) manner. While legacy PON systems were primarily designed for residential best-effort (BE) traffic, recent and emerging generations such as the recently commercialized 50G-PON and next-generation Very High Speed PON (VHSP) systems [2] aim to support heterogeneous service requirements, including

industrial and enterprise applications.

In this context, next-generation industrial networks supporting Industry 4.0+ and data-driven industrial AI applications typically generate periodic or pseudo-periodic traffic, with periodicities ranging from a few 100 μ s to tens of ms, and with highly diverse latency and jitter requirements. These requirements can be broadly categorized as 1) Extreme (<100 μ s latency, <1 μ s jitter), 2) Low (<1 ms latency, <10 μ s jitter), and 3) Relaxed (<5 ms latency, <few 100 μ s jitter). Such traffic must coexist with conventional BE FTTH services within the same access network, as illustrated in Fig. 1-a). To enable coexistence of such traffic profiles, the IEEE 802.1 Ethernet standard has been extended with Time-Sensitive Networking (TSN) features (e.g., IEEE 802.1Qbv, IEEE 802.1Qci) [3]. However, a direct transfer of these concepts to the TDM-PON US is challenging due to the reactive, traffic-driven nature of US scheduling and the inherent TDMA operation.

Latency and jitter can be reduced by tuning DBA parameters, for example by provisioning T-CONTs with committed information rates and delay tolerances. However, even with these adjustments, true determinism for Extreme or Low latency and jitter requirements is not guaranteed, as conventional bandwidth mappers (BWmappers) fail to precisely align US bursts with the application traffic. Alternatively, shorter and more frequent fixed burst allocations can improve

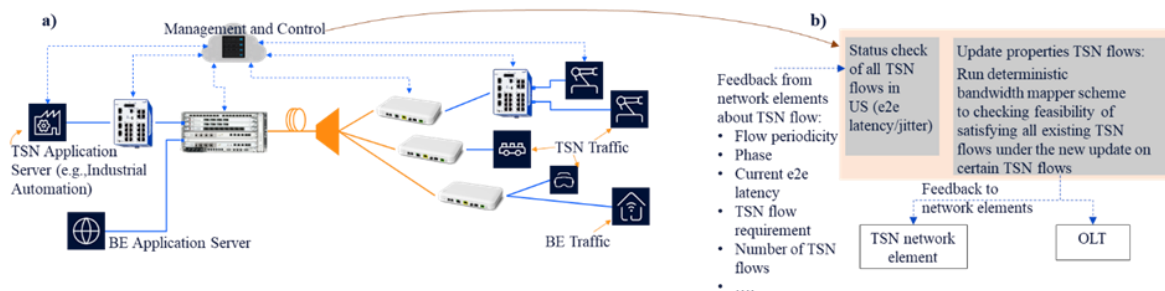


Fig. 1: a) System architecture and experimental setup of TSN applications over TDM-PON coexisting with traditional FTTH applications (incl. BE traffic). b) Illustration of the functionality of management and control in the proposed deterministic

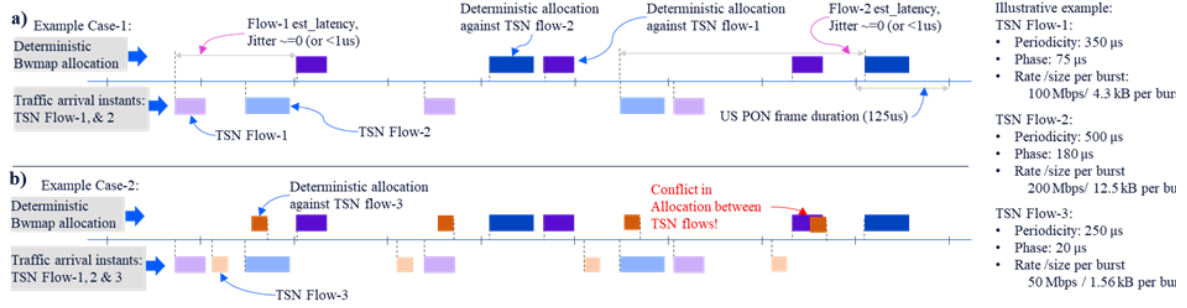


Fig. 2: Deterministic BWmap allocation of TSN flows based on knowledge of application traffic (idealistic situation) [5,6]. b) Illustrative example of challenges in allocating deterministic US bandwidth map of multiple TSN flows arising due to allocation conflicts.

latency and jitter, but at the cost of reduced bandwidth efficiency due to increased burst overhead. More promising approaches are Time-Aware DBA [4] and deterministic DBA [5], which exploit knowledge of application traffic characteristics exchanged via a cloud management and control framework (Fig. 1-b). These approaches compute deterministic BWmaps that align US bursts with the application traffic, as illustrated in Fig. 2-a). Deterministic DBA can achieve ultra-low latency ($<100 \mu$ s) and end-to-end jitter ($<1 \mu$ s) for a low to moderate number of TSN flows [5]- [6].

However, the quality of deterministic DBA scheduling scales unfavorably with the number of TSN flows, as the number of conflict situations where US burst allocations overlap within the PON frame grows exponentially. Figure 2-b) illustrates such a conflict between Flow-1 (purple) and Flow-3 (orange). Resolving all conflicts without violating per-flow latency and jitter requirements makes deterministic scheduling an increasingly difficult NP-hard problem and therefore computationally prohibitive for real-time OLT implementations.

In this paper, we introduce a deterministic BWmapper that efficiently supports a large number of time-sensitive industrial AI flows alongside best-effort traffic under the strict real-time constraints of practical OLT implementations. TSN flow requirements are directly translated into an

executable bandwidth-mapping process with linear computational complexity, bounded iterations, and localized conflict handling, ensuring predictable and scalable real-time implementation in commercial PON systems.

Deterministic bandwidth-mapper scheme

Our proposed deterministic bandwidth-mapper scheme is shown as a flow diagram in the right-hand part of Fig. 3. The scheme processes I TSN flows, sorted by their most stringent jitter requirements. The total scheduling length is the least common multiple of all flow periodicities. The first flow's burst periodic sequence is allocated across this length while accounting for its maximum packet arrival latency, thus avoiding initial conflicts, as shown in Step-1 for Flow-1 in the left-hand part of Fig. 3. For subsequent flows, allocating burst sequences may introduce conflicts. When conflicts arise, a left shift is first applied to individual conflicting bursts, as shown in Step-2 for Flow-2. If this resolves the conflict while meeting jitter requirements, it is accepted; otherwise, the entire burst periodic sequence is left-shifted as shown in Step-4 for Flow-3, and conflicts are re-assessed. This iterative process is repeated for each flow, up to J times, until all burst and jitter conflicts are resolved. A feasible burst allocation is output if all conflicts are resolved; otherwise, an infeasible allocation flag is raised. After TSN burst allocation, BE flows are inserted in a round-robin fashion into the first available empty

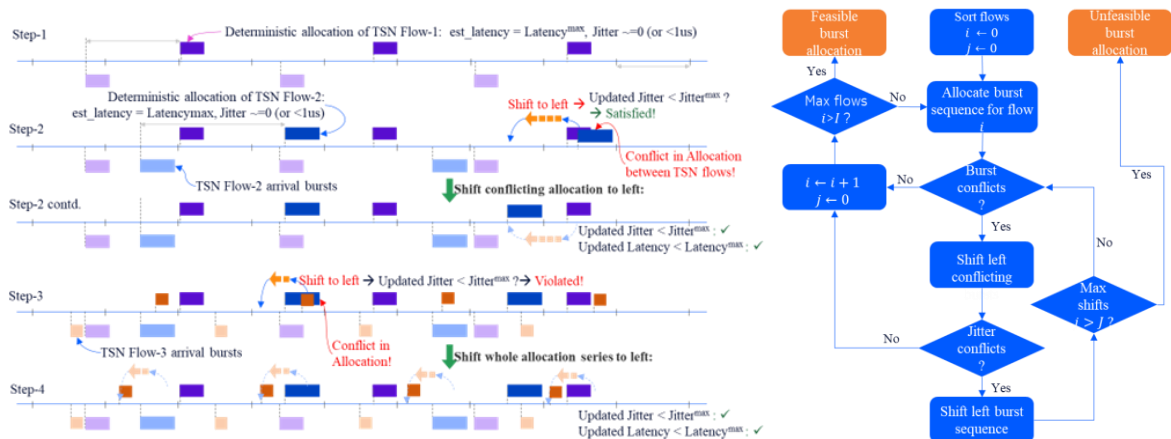


Fig. 3: Deterministic BWmap allocation of TSN flows based on knowledge of application traffic (idealistic situation) [2,3]. b) Illustrative example of challenges in allocating deterministic US bandwidth map of multiple TSN flows arising due to allocation conflicts.

positions in any US frame, up to a pre-determined maximum burst size, with remaining BE data inserted into successive PON US frames. The scheme demonstrates linear complexity with respect to the number of TSN flows, enabled by initial sorting and iterative per-burst and per-sequence conflict resolution. The number of operations is upper bounded by design, enabling predictable real-time execution and scalable deployment in practical OLT systems.

Performance Evaluation by Simulation and Experiments

The performance of the proposed scheme was evaluated through both simulation and experimental validation. The scheme was implemented in a commercial symmetric XGS-PON system operating at 10 Gbps. The experimental setup (Fig. 4) consists of multiple ONUs connected to a central OLT over a TDM-PON upstream, with some ONUs interfacing to a TSN switch and transporting both TSN and BE traffic flows. The OLT performs real-time BWmap generation and upstream scheduling, enforcing deterministic burst allocations for TSN flows while accommodating BE traffic. A traffic generator and analyzer is connected on the network side of the OLT to emulate industrial traffic patterns, inject precise TSN flows, and measure end-to-end latency and jitter. Time synchronization across the setup is ensured using PTP, enabling accurate evaluation of deterministic performance under varying TSN and BE load conditions.

Figures 5-a) and 5-b) illustrate computed BWmaps through simulation, for eight US frames under two distinct scenarios: a lightly loaded system (10% utilization) with 4 TSN and 4 BE flows, and a heavily loaded system (>75% utilization) with 6 TSN and 14 BE flows. These BWmaps are represented by colored boxes, with their horizontal width indicating the size of the bursts. Figure 5-c) compares the computational complexity against a brute-force approach, demonstrating the linear complexity of the

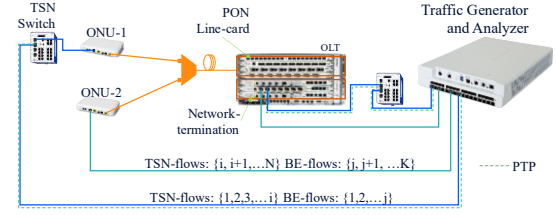


Fig. 4: Experimental Setup for performance evaluation of our proposed Deterministic BWmapper scheme

proposed scheme, which makes it suitable for real-time BWmap generation. Figure 5-d) presents experimental US latency and jitter performance for BWmap generation under two PON load conditions: low (2 TSN + 1 BE flow) and high (4 TSN + 2 BE flows). Bars represent latencies, while bounding intervals indicate jitter. Notably, the proposed scheme achieves the deterministic sub-ms end-to-end latency targets and μ s-level jitter targets for TSN flows even under high PON loads. Only BE traffic experiences increased latency and jitter with higher traffic load.

Conclusions

This paper presented a deterministic BWmapper for US TDM-PON that enables scalable and real-time support of time-sensitive traffic alongside BE traffic, targeting the stringent communication requirements of industrial AI applications. Beyond algorithmic design, the key contribution lies in the practical realization and validation of the proposed BWmapper in a real XGS-PON system, demonstrating real-time generation of TSN-aware deterministic BWmaps that adapt to multiple concurrent flows. Experimental results confirm that the scheme achieves sub-millisecond end-to-end latency and microsecond-level jitter for multiple TSN flows, even under high US load, while allowing BE traffic to coexist efficiently. These results demonstrate that deterministic networking support for emerging time-sensitive industrial AI applications can be achieved at scale in operational PON systems without sacrificing real-time feasibility or bandwidth efficiency.

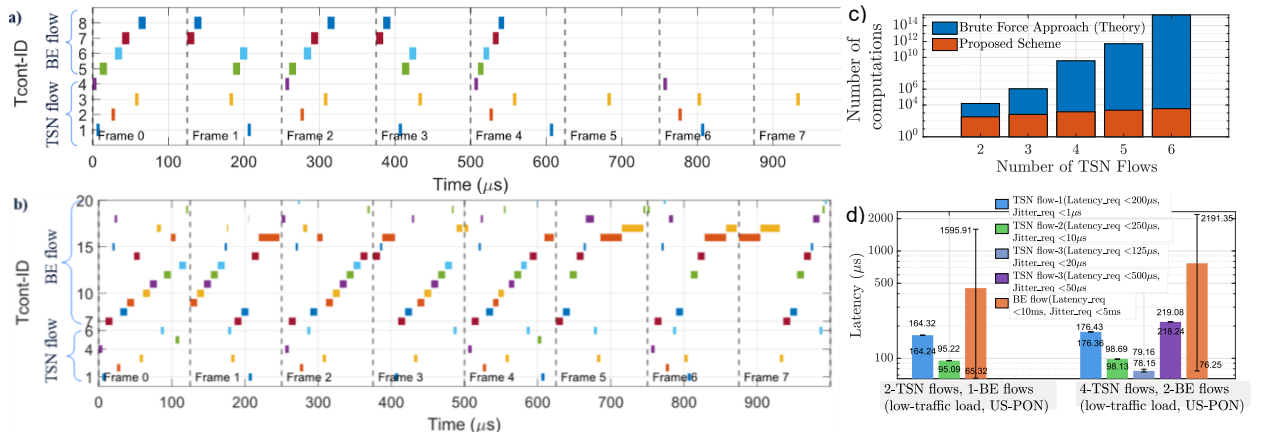


Fig. 5. Generation of deterministic bwmap (a) for lightly loaded PON (~10%) (b) for highly loaded PON (>75%) (c) Time complexity for the proposed deterministic bwmap allocation algorithm in-comparison to a brute force approach. (d) Latency and jitter performances corresponding to the bwmap generation under low (<10%) and high (>50%) PON loads.

Acknowledgements

This work was partially funded by the Federal Ministry of Education and Research (BMBF), Germany. SUSTAINET-iNnOvAte: 16KIS2255K, and the Flemish Government funding agency VLAIO (project no. HBC.2024.0499).

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